Iridium

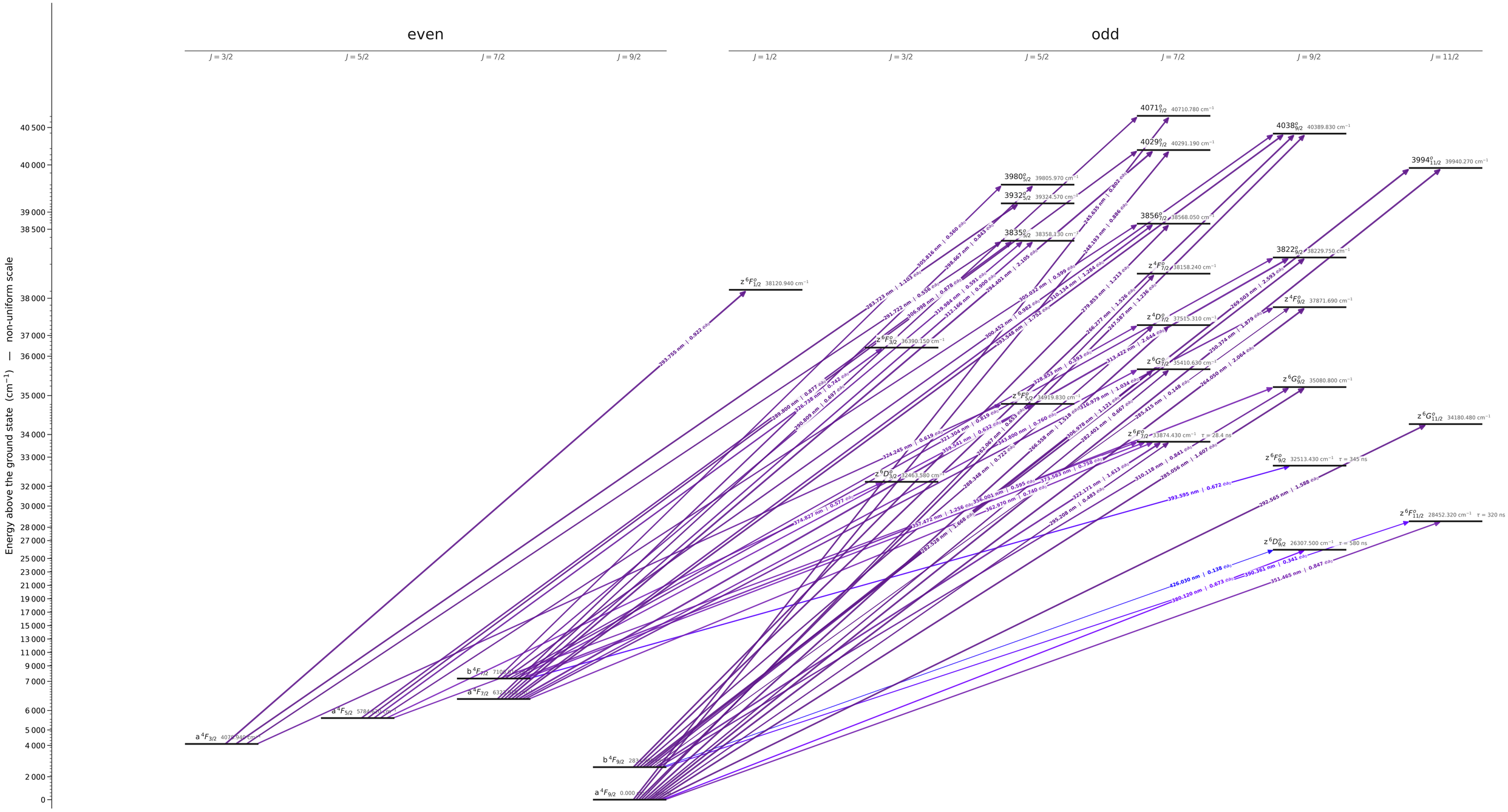
Ir (Ir I). Strongest classified transitions below 2 μm (NIST): 29 levels, 53 lines.

Ir^{+} ionisation limit 72323.90 cm^{-1} = 8.96702 eV (138.267 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
a $^4F_{9/2}$ – 4038°$_{9/2}$	247.5871	247.5123	1210.8566	1.236	2.040e+07	–	–	first-step line of the in-gas-cell laser resonance ioniza...
a $^4F_{9/2}$ – z $^6F°_{11/2}$	351.4652	351.3647	852.9791	0.847	2.790e+06	0.4974	89.2	351 nm line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{193}Ir hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a $^4F_{9/2}$	62.655370(20)	426.23550(20)	F=3: -321.55 F=4: -355.08 F=5: -130.61 F=6: +529.48
z $^6F°_{11/2}$	308(4)	-750(30)	F=4: -3313.23 F=5: -1295.95 F=6: +674.77 F=7: +2353.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
a $^4F_{9/2}$ – 4038°$_{9/2}$	247.587	193-191	-8100.00
a $^4F_{9/2}$ – z $^6F°_{11/2}$	351.465	193-191	-2000(10)

Shift = ν(first isotope) − ν(second isotope).